

Expt 1. Study of Network devices in detail and connect the computers in Local Area Network.

Aim:

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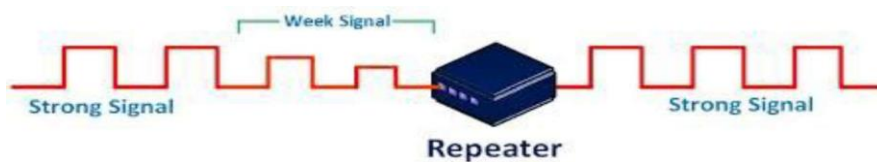
Theory:

a) Study of Network Devices in Detail

- Repeater
- Hub
- Switch
- Bridge
- Router
- GateWay

1.Repeater:

A repeater receives a weak signal, regenerates it at higher power, and retransmits it to extend the communication distance. It has two ports and connects two network segments.



2. Hub:

An Ethernet Hub is a multiport repeater that connects multiple Ethernet devices into a single network segment. It operates at the Physical Layer (Layer 1) of the OSI model. It receives and regenerates signals, then broadcasts them to all connected ports. When a collision occurs, the hub sends a jam signal to all devices, which then retransmit after a random backoff period.

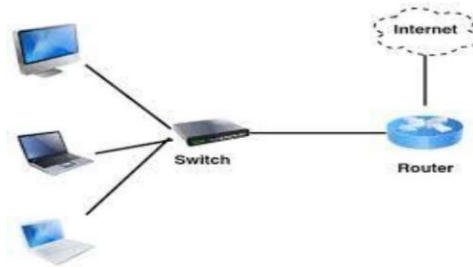


3. Switch :

A network switch connects multiple devices or network segments and mainly operates at the Data Link Layer (Layer 2) of the OSI model. It examines the destination MAC address of each frame and forwards it only to the appropriate port, reducing unnecessary traffic and improving network efficiency. Layer 3 switches can also perform routing functions for larger networks.

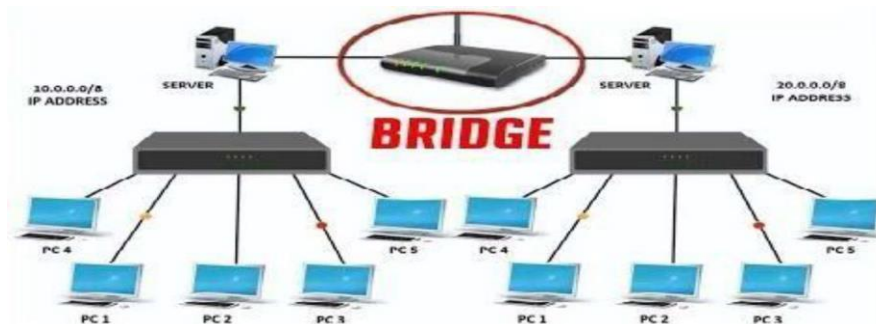


Switch



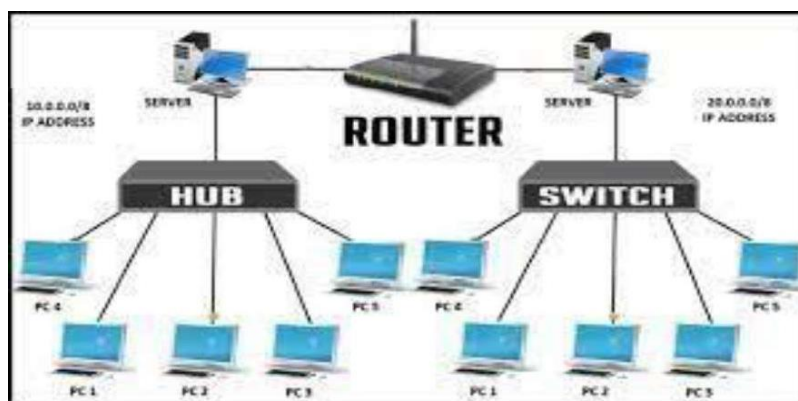
4. Bridge:

A network bridge connects multiple network segments and operates at the Data Link Layer (Layer 2) of the OSI model. It examines incoming frames and forwards or filters them based on MAC addresses, reducing unnecessary traffic and improving network performance. A switch is essentially a multiport bridge, so the terms are often used interchangeably.



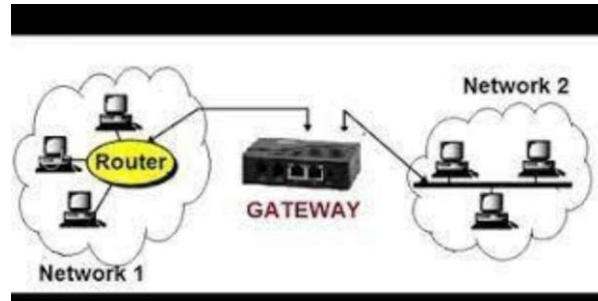
5. Router:

A router connects two or more networks and operates at the Network Layer (Layer 3) of the OSI model. It examines the destination IP address of packets and forwards them through the appropriate network. Routers use routing tables to determine the best path for delivering packets between networks.



6. Gateway:

A gateway connects two or more networks that use different communication protocols and enables them to communicate by performing protocol conversion. It ensures interoperability between dissimilar networks and may include protocol translators, rate converters, or signal translators. A protocol gateway specifically converts one network protocol into another for effective communication.



Aim

To connect computers in a Local Area Network (LAN) and share the Internet connection from the host computer with client computers.

Procedure

On the Host Computer

Follow these steps to enable Internet Connection Sharing (ICS):

1. Log on to the host computer as an **Administrator** or **Owner**.
2. Click **Start**, and then click **Control Panel**.
3. Click **Network and Internet Connections**.
4. Click **Network Connections**.
5. Right-click the Internet connection you want to share (for example, a dial-up, broadband, or other network connection), and then click **Properties**.
6. Click the **Advanced** tab.
7. Under **Internet Connection Sharing**, select the **Allow other network users to connect through this computer's Internet connection** check box.
8. If you are sharing a dial-up Internet connection, select the **Establish a dial-up connection whenever a computer on my network attempts to access the Internet** check box, if required.
9. Click **OK**.

A confirmation message appears:

When Internet Connection Sharing is enabled, your LAN adapter will be set to use the IP address 192.168.0.1. Your computer may lose connectivity with other computers on your network. If these other computers have static IP addresses, it is a good idea to set them to

obtain their IP addresses automatically. Are you sure you want to enable Internet Connection Sharing?

10. Click **Yes**.

The Internet connection is now shared with other computers on the Local Area Network (LAN). The LAN adapter on the host computer is automatically assigned:

- **IP Address:** 192.168.0.1
- **Subnet Mask:** 255.255.255.0

On the Client Computer

To access the Internet through the shared connection:

1. Ensure that the LAN adapter is connected to the host computer.
2. Open **Control Panel** → **Network and Internet Connections** → **Network Connections**.
3. Right-click **Local Area Connection** and select **Properties**.
4. Select **Internet Protocol (TCP/IP)** or **Internet Protocol Version 4 (TCP/IPv4)**, and click **Properties**.
5. Select **Obtain an IP address automatically**.
6. Select **Obtain DNS server address automatically**.
7. Click **OK**, and then close all dialog boxes.
8. Restart the computer if prompted.

The client computer will automatically receive an IP address from the host computer and will be able to access the Internet through the shared connection.